Computing Laboratory · Fakultas Informatika · Telkom University

Advance Study Group

Belajar AI Lewat Paper

Pertemuan 0



Pretest

8 menit

- 12 soal pendek, tanpa perangkat
- Tanpa nilai. Pakai nama samaran
- Kosongkan soal yang tidak kamu tahu
- Hasilnya hanya untuk merencanakan kelas

Semua AI bermula dari paper



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Komodo Mlipir Algorithm

Suyanto Suyanto *, Alifya Aisyah Ariyanto, Alifya Fatimah Ariyanto

School of Computing, Telkom University, Bandung, Indonesia

ARTICLE INFO

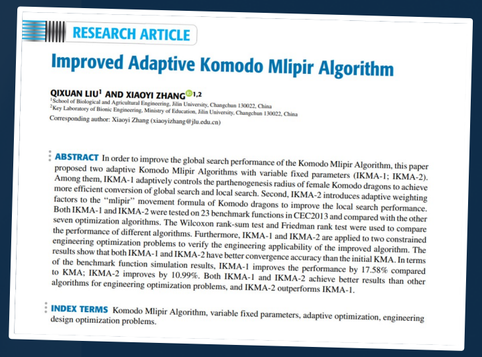
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RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QIXUAN LIU[1] AND XIAOYI ZHANG[1,2]
[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

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INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.



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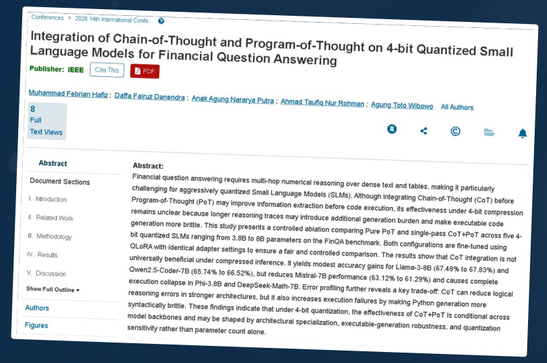
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Ahmad Daffa Aminuddin Siregar ; Kemas Muslim Lhaksmana ; Andy Maulana Yusuf All Authors

7 Full Text Views

Abstract
Document Sections
I. Introduction
II. Related Work
III. Methodology
IV. Experiments
V. Results
Show Full Outline
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Conferences > 2026 14th International Confe...

Integration of Chain-of-Thought and Program-of-Thought on 4-bit Quantized Small Language Models for Financial Question Answering

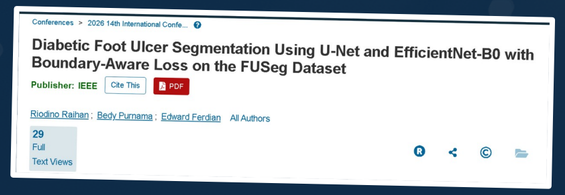
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Muhammad Febrian Hafiz ; Daffa Fairuz Danendra ; Anak Agung Narayya Putra ; Ahmad Taufiq Nur Rohman ; Agung Toto Wibowo All Authors

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Abstract
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Figures

Abstract:
Financial question answering requires multi-hop numerical reasoning over dense text and tables, making it particularly challenging for aggressively quantized Small Language Models (SLMs). Although integrating Chain-of-Thought (CoT) before Program-of-Thought (PoT) may improve information extraction before code execution, its effectiveness under 4-bit compression remains unclear because longer reasoning traces may introduce additional generation burden and make executable code generation more brittle. This study presents a controlled ablation comparing Pure PoT and single-pass CoT+PoT across five 4-bit quantized SLMs ranging from 3.8B to 8B parameters on the FinQA benchmark. Both configurations are fine-tuned using QLoRA with identical adapter settings to ensure a fair and controlled comparison. The results show that CoT integration is not universally beneficial under compressed inference. It yields modest accuracy gains for Llama-3-8B (67.48% to 67.83%) and Qwen2.5-Coder-7B (65.74% to 66.52%), but reduces Mistral-7B performance (63.12% to 61.29%) and causes complete execution collapse in Phi-3.8B and DeepSeek-Math-7B. Error profiling further reveals a key trade-off: CoT can reduce logical reasoning errors in stronger architectures, but it also increases execution failures by making Python generation more syntactically brittle. These findings indicate that under 4-bit quantization, the effectiveness of CoT+PoT is conditional across model backbones and may be shaped by architectural specialization, executable-generation robustness, and quantization sensitivity rather than parameter count alone.



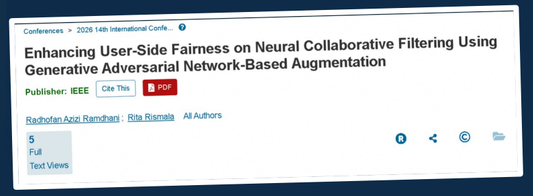
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Diabetic Foot Ulcer Segmentation Using U-Net and EfficientNet-B0 with Boundary-Aware Loss on the FUSeg Dataset

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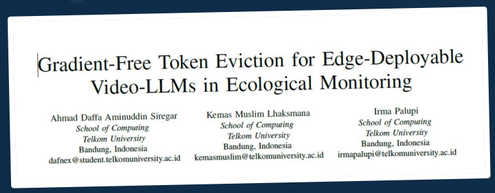
Radhofan Azizi Ramdhani ; Rita Rismala All Authors

5 Full Text Views



Reference-vector competitor pairing for constrained large-scale multi-objective competitive swarm optimisation

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Gradient-Free Token Eviction for Edge-Deployable Video-LLMs in Ecological Monitoring

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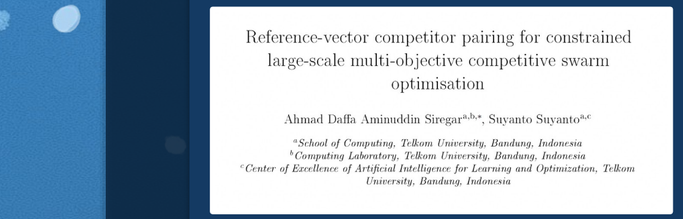
Hari ini kita belajar membaca dunia ini

Agenda Hari Ini

0:00	Pretest (8 menit)
0:08	Dinding paper dan agenda (5)
0:13	Empat jenis paper (30)
0:43	Cara membaca paper (10)
0:53	Detektif Paper (25)
1:18	Istirahat (5)
1:23	Dari paper ke kelas ini (8)
1:31	Cara kerja kelas (15)
1:46	Stasiun persiapan (10)
1:56	Tiket keluar (4)

Empat Jenis Paper

Riset murni



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Ide atau algoritma baru, diuji pada soal standar

Riset terapan



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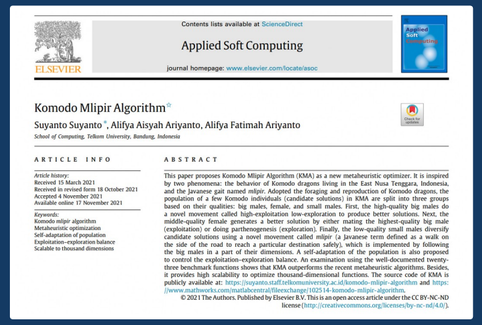
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AI dipakai atau diaudit pada masalah nyata

Bidang berkembang



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Satu ide diperbaiki dan dipakai orang lain

Dari paper ke poster



21

Gradient-Free Token Eviction for Edge-Deployable Video-LLMs in Ecological Monitoring

Motivation: the edge bottleneck

Key idea

Method

Results

19.1%

52%

4.5 GB

37

Takeaways

Limitations

Hasil riset dikomunikasikan di konferensi

RISET MURNI

Ide baru untuk sebuah algoritma

Reference-vector competitor pairing for constrained large-scale multi-objective competitive swarm optimisation

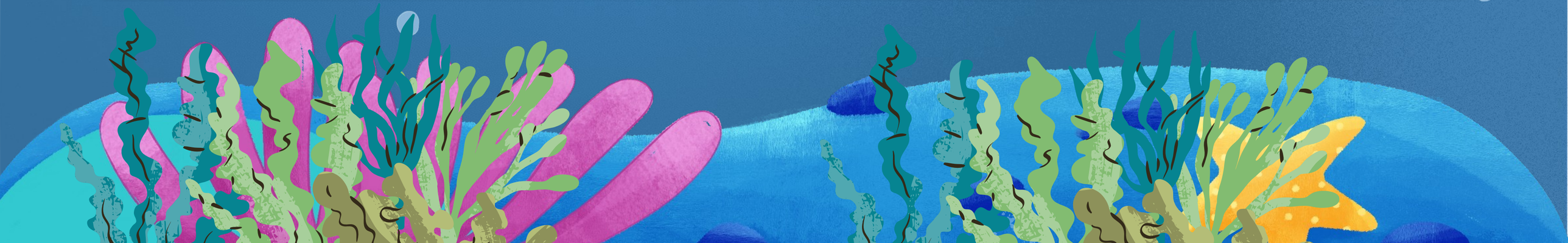
Ahmad Daffa Aminuddin Siregar[a,b,*], Suyanto Suyanto[a,c]

[a] *School of Computing, Telkom University, Bandung, Indonesia*
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Siregar dan Suyanto, Telkom University
Computing Laboratory, Telkom University
Pusat Keunggulan AI untuk Pembelajaran dan Optimasi

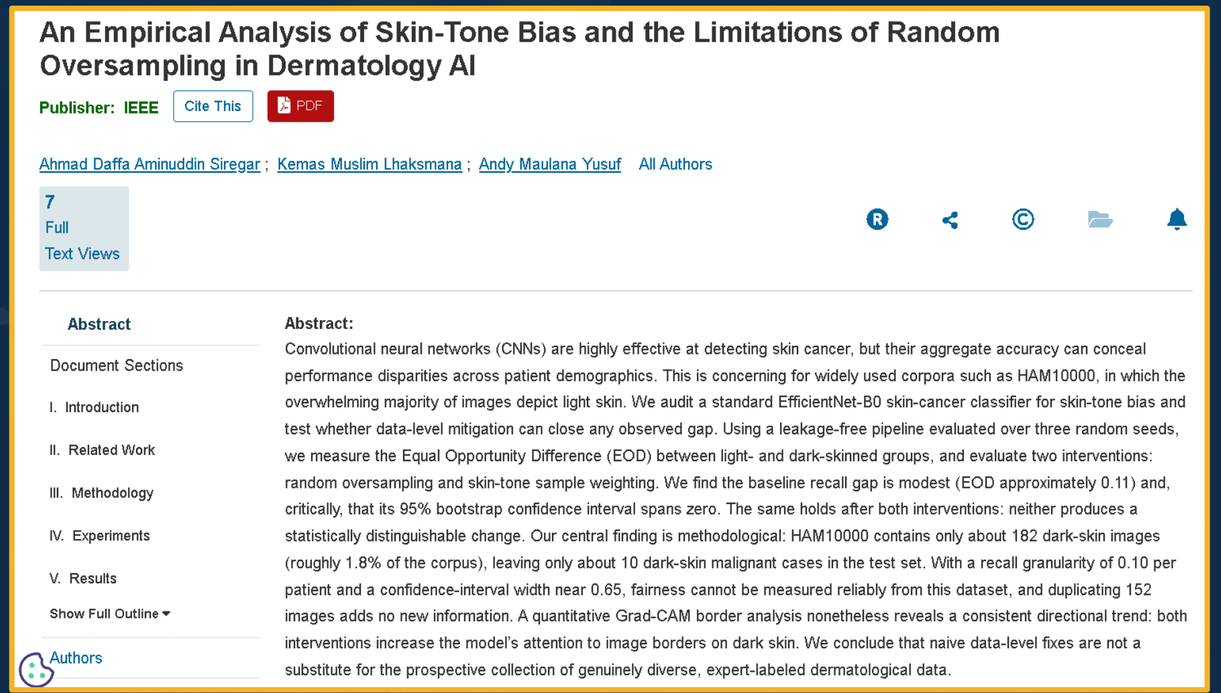
Bedah judulnya

- **Reference-vector competitor pairing**
  ide barunya
- **constrained**
  ada batasan
- **large-scale**
  variabel sangat banyak
- **multi-objective**
  beberapa tujuan sekaligus
- **competitive swarm optimisation**
  algoritma dasar yang diperbaiki



RISET TERAPAN

Mengaudit AI pada masalah nyata



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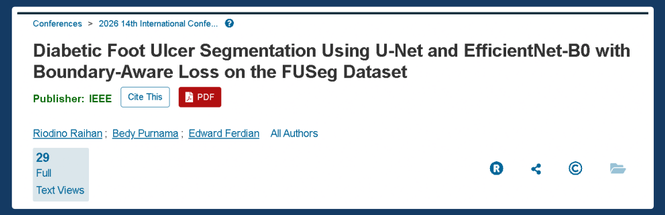
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- Model: EfficientNet-B0 untuk kanker kulit
- Data HAM10000: hanya sekitar 182 gambar kulit gelap (1,8%)
- Selisih recall (EOD) sekitar 0,11, tetapi selang kepercayaan 95% mencakup nol
- **Kesimpulan: perbaikan data saja tidak cukup**

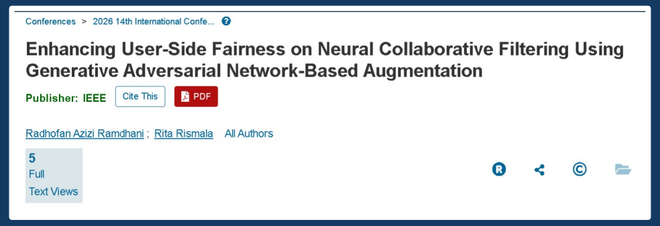
RISET TERAPAN

Satu pola, banyak bidang



Citra medis

Segmentasi luka kaki diabetik dengan U-Net dan EfficientNet-B0



Sistem rekomendasi

Keadilan pada rekomendasi, dengan augmentasi berbasis GAN



Model bahasa

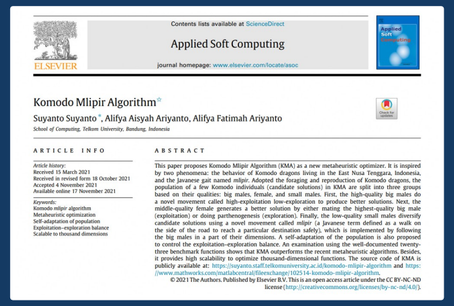
Penalaran soal keuangan pada model kecil 4-bit

Hasil campuran pun layak terbit: CoT menaikkan Llama-3-8B, tetapi menurunkan Mistral-7B

BIDANG BERKEMBANG

Satu ide, tiga paper

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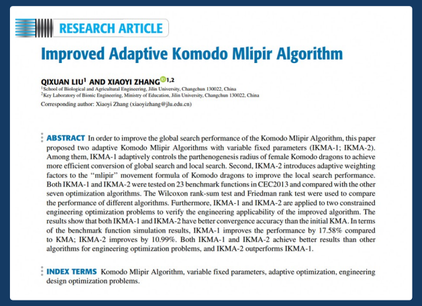
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Paper asli

Applied Soft Computing

Telkom University

2022



RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QIXUAN LIU[1] AND XIAOYI ZHANG[2,1,2]

[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China

Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

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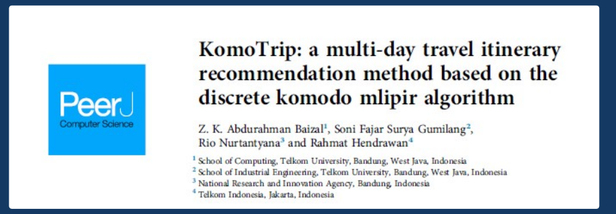
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Diperbaiki orang lain

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PeerJ Computer Science

KomoTrip: a multi-day travel itinerary recommendation method based on the discrete komodo mlipir algorithm

Z. K. Abdurahman Baizal[1], Soni Fajar Surya Gumilang[2], Rio Nurtantyana[3] and Rahmat Hendrawan[4]

[1] School of Computing, Telkom University, Bandung, West Java, Indonesia
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[3] National Research and Innovation Agency, Bandung, Indonesia
[4] Telkom Indonesia, Jakarta, Indonesia

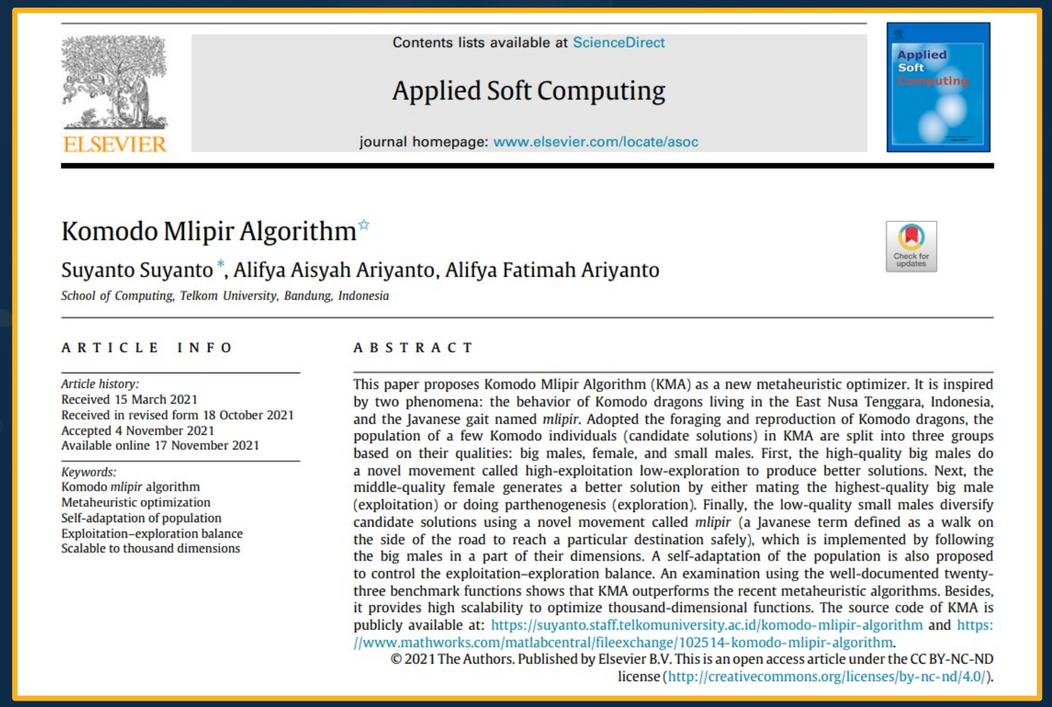
Dipakai untuk masalah nyata

PeerJ Computer Science

Telkom University dan mitra

BIDANG BERKEMBANG

Paper asli: Komodo Mlipir Algorithm



Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Komodo Mlipir Algorithm☆

Suyanto Suyanto *, Alifya Aisyah Ariyanto, Alifya Fatimah Ariyanto

School of Computing, Telkom University, Bandung, Indonesia

ARTICLE INFO

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Keywords:
Komodo *mlipir* algorithm
Metaheuristic optimization
Self-adaptation of population
Exploitation–exploration balance
Scalable to thousand dimensions

ABSTRACT

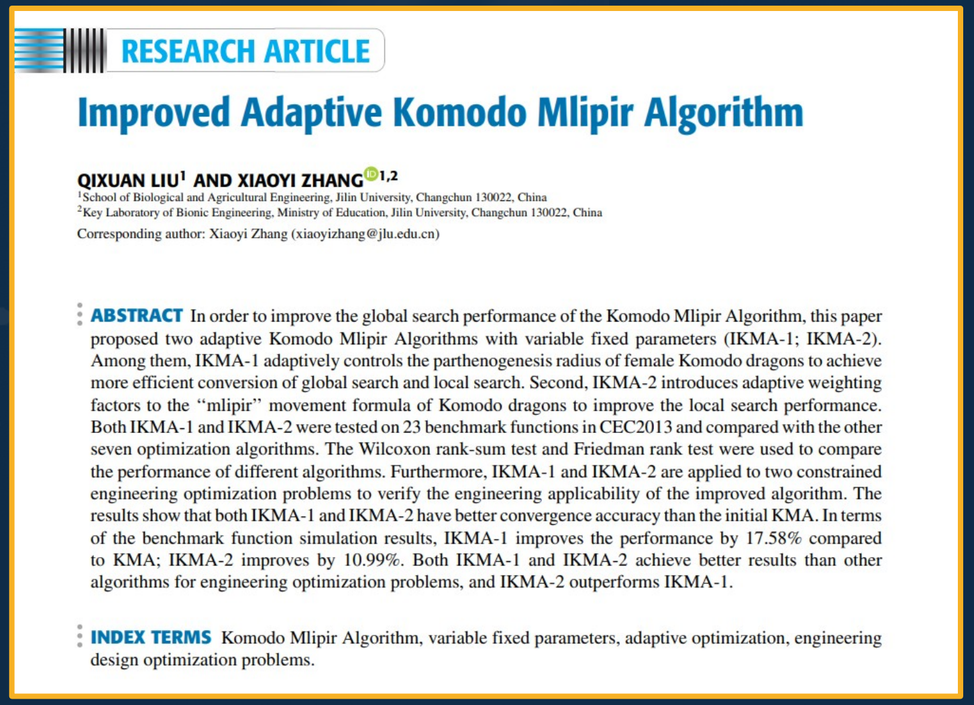
This paper proposes Komodo Mlipir Algorithm (KMA) as a new metaheuristic optimizer. It is inspired by two phenomena: the behavior of Komodo dragons living in the East Nusa Tenggara, Indonesia, and the Javanese gait named *mlipir*. Adopted the foraging and reproduction of Komodo dragons, the population of a few Komodo individuals (candidate solutions) in KMA are split into three groups based on their qualities: big males, female, and small males. First, the high-quality big males do a novel movement called high-exploitation low-exploration to produce better solutions. Next, the middle-quality female generates a better solution by either mating the highest-quality big male (exploitation) or doing parthenogenesis (exploration). Finally, the low-quality small males diversify candidate solutions using a novel movement called *mlipir* (a Javanese term defined as a walk on the side of the road to reach a particular destination safely), which is implemented by following the big males in a part of their dimensions. A self-adaptation of the population is also proposed to control the exploitation–exploration balance. An examination using the well-documented twenty-three benchmark functions shows that KMA outperforms the recent metaheuristic algorithms. Besides, it provides high scalability to optimize thousand-dimensional functions. The source code of KMA is publicly available at: https://suyanto.staff.telkomuniversity.ac.id/komodo-mlipir-algorithm and https://www.mathworks.com/matlabcentral/fileexchange/102514-komodo-mlipir-algorithm.

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- Terinspirasi komodo di NTT dan gaya jalan mlipir
- Populasi dibagi tiga: jantan besar, betina, jantan kecil
- Diuji pada 23 fungsi benchmark
- Kode sumber dibuka untuk umum
- **Open access, lahir di School of Computing, Telkom University**

BIDANG BERKEMBANG

Diperbaiki: Improved Adaptive KMA



RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QIXUAN LIU[1] AND XIAOYI ZHANG[1,2]
[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

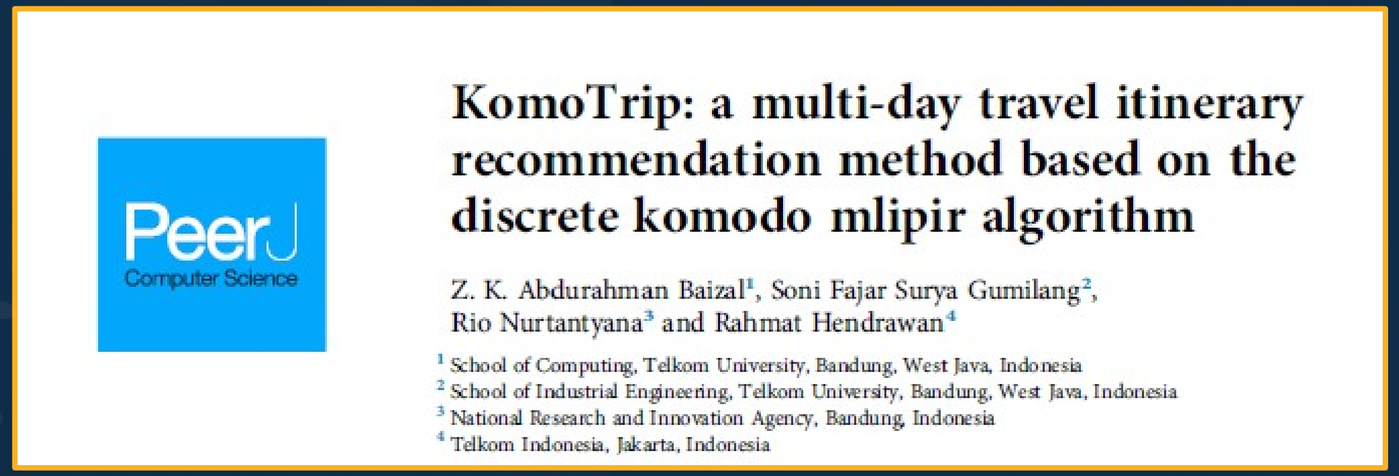
ABSTRACT In order to improve the global search performance of the Komodo Mlipir Algorithm, this paper proposed two adaptive Komodo Mlipir Algorithms with variable fixed parameters (IKMA-1; IKMA-2). Among them, IKMA-1 adaptively controls the parthenogenesis radius of female Komodo dragons to achieve more efficient conversion of global search and local search. Second, IKMA-2 introduces adaptive weighting factors to the "mlipir" movement formula of Komodo dragons to improve the local search performance. Both IKMA-1 and IKMA-2 were tested on 23 benchmark functions in CEC2013 and compared with the other seven optimization algorithms. The Wilcoxon rank-sum test and Friedman rank test were used to compare the performance of different algorithms. Furthermore, IKMA-1 and IKMA-2 are applied to two constrained engineering optimization problems to verify the engineering applicability of the improved algorithm. The results show that both IKMA-1 and IKMA-2 have better convergence accuracy than the initial KMA. In terms of the benchmark function simulation results, IKMA-1 improves the performance by 17.58% compared to KMA; IKMA-2 improves by 10.99%. Both IKMA-1 and IKMA-2 achieve better results than other algorithms for engineering optimization problems, and IKMA-2 outperforms IKMA-1.

INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.

- Dua varian: IKMA-1 dan IKMA-2
- **Diklaim memperbaiki KMA: 17,58% (IKMA-1) dan 10,99% (IKMA-2)**
- Diuji pada 23 fungsi CEC2013 dan dua masalah rekayasa
- Dibandingkan dengan tujuh algoritma lain, dengan uji Wilcoxon dan Friedman
- Ditulis oleh peneliti Jilin University, Tiongkok

BIDANG BERKEMBANG

Dipakai: KomoTrip



PeerJ Computer Science

KomoTrip: a multi-day travel itinerary recommendation method based on the discrete komodo mlipir algorithm

Z. K. Abdurahman Baizal[1], Soni Fajar Surya Gumilang[2], Rio Nurtantyana[3] and Rahmat Hendrawan[4]

[1] School of Computing, Telkom University, Bandung, West Java, Indonesia
[2] School of Industrial Engineering, Telkom University, Bandung, West Java, Indonesia
[3] National Research and Innovation Agency, Bandung, Indonesia
[4] Telkom Indonesia, Jakarta, Indonesia

Masalah nyata

Menyusun itinerari wisata beberapa hari

Metode

KMA versi diskret

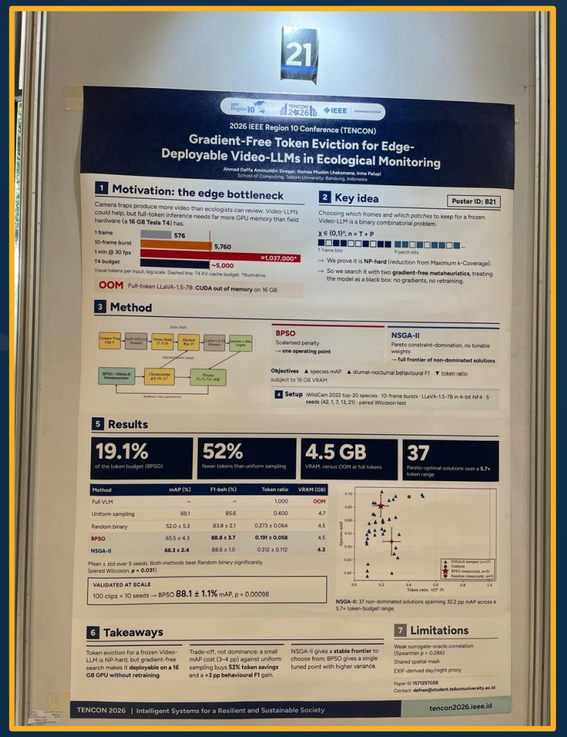
Judul sudah memberi tahu keduanya

Telkom University, BRIN, dan Telkom Indonesia

PeerJ Computer Science, 12 November 2025

DARI PAPER KE POSTER

Hasil riset di konferensi



Gradient-Free Token Eviction for Edge-Deployable Video-LLMs in Ecological Monitoring

Ahmad Daffa Aminuddin Siregar
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Satu paper, satu poster di IEEE TENCON 2026

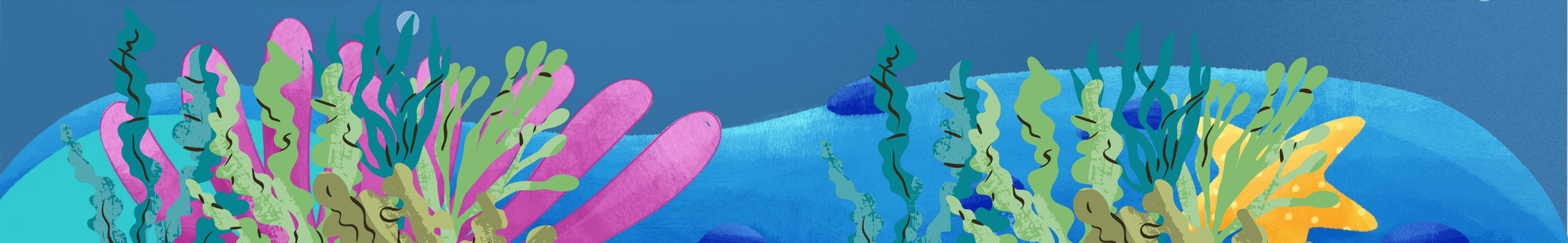
Tujuh kotak: Motivasi, Ide kunci, Metode, Setup, Hasil, Takeaways, Limitations

Angka utama dibuat besar: 19,1% budget token, 52% lebih sedikit token, 4,5 GB VRAM

DARI PAPER KE KELAS

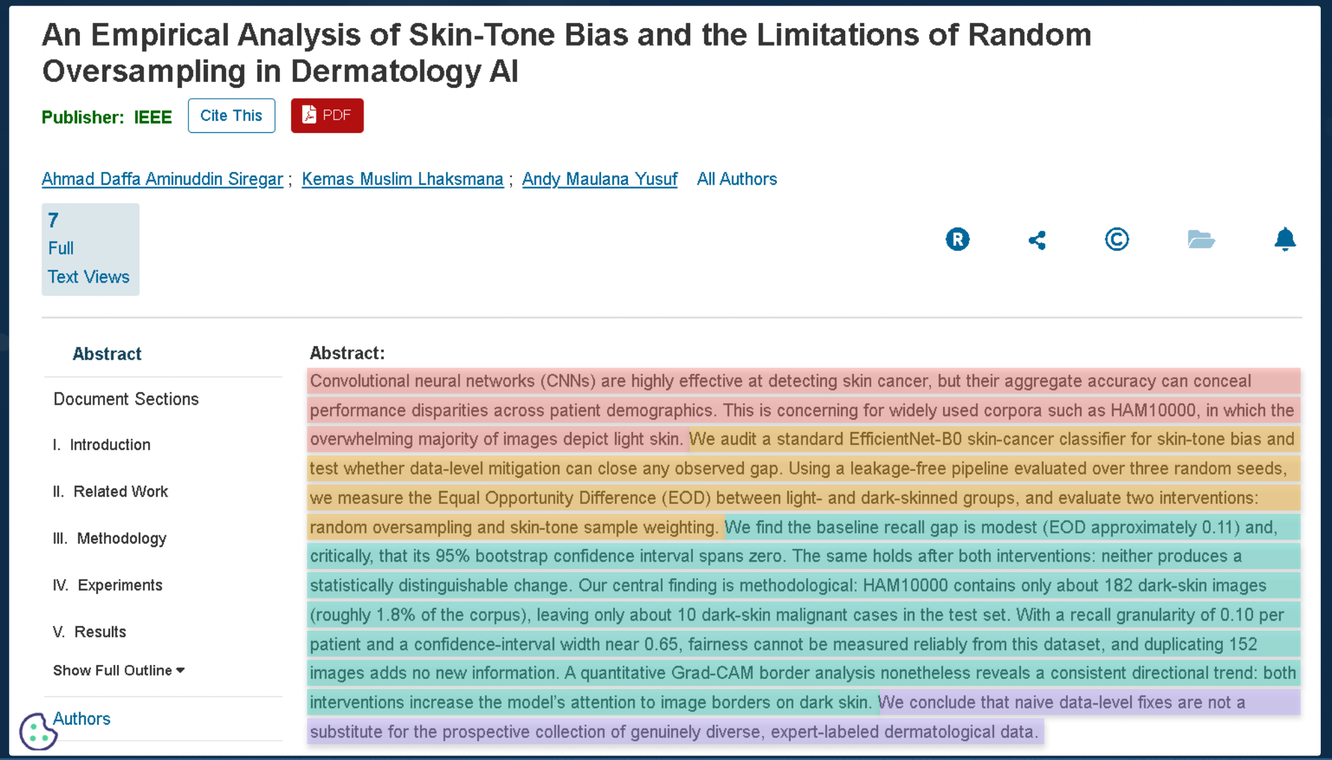
Poster ini memakai isi kelas kita

Di poster	Di kelas ini
Pilih frame dan patch mana yang disimpan: soal biner	Representasi solusi (Pekan 11)
BPSO: binary particle swarm	Metaheuristik populasi, PSO (Pekan 10)
NSGA-II: banyak tujuan, solusi Pareto	Multi-objektif (topik lanjutan, opsional)
Gradient-free: model dianggap kotak hitam	Optimasi tanpa gradien (Pekan 9 sampai 12)
5 seed, rata-rata ± simpangan baku, uji Wilcoxon	Protokol eksperimen (Pekan 9 dan proyek akhir)
Video-LLM beku, LLaVA-1.5-7B 4-bit	Transformer dan LLM (Pekan 14)
Kotak Limitations	Catatan teknis dengan satu kasus gagal



CARA MEMBACA

Satu abstrak, empat bagian



An Empirical Analysis of Skin-Tone Bias and the Limitations of Random Oversampling in Dermatology AI

Publisher: IEEE | Cite This | PDF

Ahmad Daffa Aminuddin Siregar ; Kemas Muslim Lhaksmana ; Andy Maulana Yusuf All Authors

7 Full Text Views

Abstract

Document Sections

I. Introduction

II. Related Work

III. Methodology

IV. Experiments

V. Results

Show Full Outline

Authors

Abstract:

Convolutional neural networks (CNNs) are highly effective at detecting skin cancer, but their aggregate accuracy can conceal performance disparities across patient demographics. This is concerning for widely used corpora such as HAM10000, in which the overwhelming majority of images depict light skin. We audit a standard EfficientNet-B0 skin-cancer classifier for skin-tone bias and test whether data-level mitigation can close any observed gap. Using a leakage-free pipeline evaluated over three random seeds, we measure the Equal Opportunity Difference (EOD) between light- and dark-skinned groups, and evaluate two interventions: random oversampling and skin-tone sample weighting. We find the baseline recall gap is modest (EOD approximately 0.11) and, critically, that its 95% bootstrap confidence interval spans zero. The same holds after both interventions: neither produces a statistically distinguishable change. Our central finding is methodological: HAM10000 contains only about 182 dark-skin images (roughly 1.8% of the corpus), leaving only about 10 dark-skin malignant cases in the test set. With a recall granularity of 0.10 per patient and a confidence-interval width near 0.65, fairness cannot be measured reliably from this dataset, and duplicating 152 images adds no new information. A quantitative Grad-CAM border analysis nonetheless reveals a consistent directional trend: both interventions increase the model's attention to image borders on dark skin. We conclude that naive data-level fixes are not a substitute for the prospective collection of genuinely diverse, expert-labeled dermatological data.

Baca abstrak seperti ini

Masalah
Apa yang bermasalah?

Metode
Apa yang dikerjakan?

Hasil
Apa temuannya? Angkanya?

Kesimpulan
Apa artinya, dan batasnya?

Bagian paper: Pendahuluan, Related Work, Metodologi, Eksperimen, Hasil

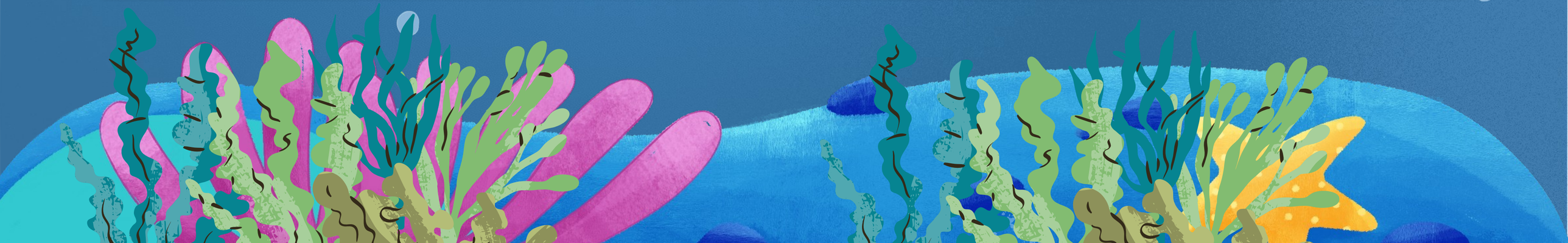
Membaca Paper dalam 10 Menit

Urutan membaca

1. Judul
2. Abstrak
3. Gambar dan tabel
4. Kesimpulan dan keterbatasan
5. Metode, hanya jika perlu

Lima pertanyaan

1. Masalah apa yang dikerjakan?
2. Apa yang diusulkan atau dilakukan?
3. Bagaimana diuji?
4. Apa angka kuncinya?
5. Apa keterbatasannya?



Detektif Paper

25 menit · berpasangan

1. Ambil satu kartu paper (A sampai E)
2. Jawab lima pertanyaan di kartu, 12 menit
3. Tandai angka kunci dan satu keterbatasan
4. Bagikan hasilnya: satu menit per paper


An Empirical Analysis of Skin-Tone Bias and the Limitations of Random Oversampling in Dermatology AI

Publisher: IEEE

Abstract

Document Sections

I. Introduction

II. Related Work

III. Methodology

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A


Integration of Chain-of-Thought and Program-of-Thought on 4-bit Quantized Small Language Models for Financial Question Answering

Publisher: IEEE

Abstract

Document Sections

I. Introduction

II. Related Work

III. Methodology

IV. Results

V. Discussion

Show Full Outline

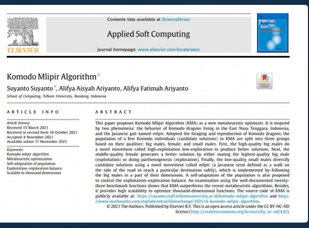
Authors

Figures

Abstract:

Financial question answering requires multi-hop numerical reasoning over dense text and tables, making it particularly challenging for aggressively quantized Small Language Models (SLMs). Although integrating Chain-of-Thought (CoT) before Program-of-Thought (PoT) may improve information extraction before code execution, its effectiveness under 4-bit compression remains unclear because longer reasoning traces may introduce additional generation burden and make executable code generation more brittle. This study presents a controlled ablation comparing Pure PoT and single-pass CoT+PoT across five 4-bit quantized SLMs ranging from 1.88 to 8B parameters on the FinQA benchmark. Both configurations are fine-tuned using QLoRA with identical adapter settings to ensure a fair and controlled comparison. The results show that CoT integration is not universally beneficial under compressed inference. A yields modest accuracy gains for Llama-3.1B (67.40% to 67.87%) and Qwen2.5-Coder-7B (85.74% to 86.52%), but reduces Mistral-7B performance (0.12% to 61.29%) and causes complete execution collapse in Phi-3.8B and DeepSeek-Math-7B. Error profiling further reveals a key trade-off: CoT can reduce logical reasoning errors in stronger architectures, but it also increases execution failures by making Python generation more syntactically brittle. These findings indicate that under 4-bit quantization, the effectiveness of CoT+PoT is conditional across model backbones and may be shaped by architectural specialization, executable-generation robustness, and quantization sensitivity rather than parameter count alone.

B


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Komodo Mlipir Algorithm

Suyanto Suyanto*, Alifya Aisyah Ariyanto, Alifya Fatimah Ariyanto

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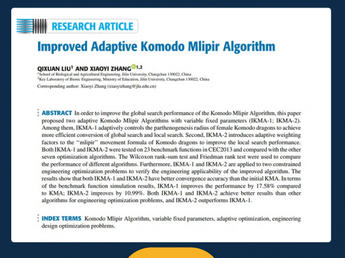
Keywords:
Komodo mlipir algorithm
Metaheuristic algorithm
Self-adaptation of population
Exploration-exploitation balance
Scalable to thousand dimensions

ABSTRACT

This paper proposes Komodo Mlipir Algorithm (KMA) as a new metaheuristic optimizer. It is inspired by two phenomena: the behavior of Komodo dragons living in the East Nusa Tenggara, Indonesia, and the Javanese gak ngenel mlipir. Adopted the foraging and reproduction of Komodo dragons, the population of a few Komodo individuals (candidate solutions) in KMA are split into three groups: male, female, and juvenile. Each group has a unique way to move. The male group is the first time a novel movement called high-exploitation low-exploration to produce better solutions. Next, the middle-quality female generates a better solution by either mating the highest-quality big male solution or moving to the best solution position. Meanwhile, the juvenile group moves around the candidate solutions using a novel movement called mlipir. Juveniles were defined as a walk on the side of the road. In such a particular denotation subtly, which is implemented by following the big males in a part of their dimensions. A self-adaptation of the population is also proposed to control the exploitation-exploration balance. An examination using the well-documented twenty-three benchmark functions shows that KMA is competitive with other well-known metaheuristics. Besides, it provides high scalability to optimize thousand-dimensional functions. The source code of KMA is publicly available at: https://suyanto.staff.telkomuniversity.ac.id/komodo-mlipir-algorithm and https://www.mathworks.com/matlabcentral/fileexchange/85731-komodo-mlipir-algorithm.

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C


RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QEXUAN LIU[1] AND XIAOYI ZHANG[2,3]

[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

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INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.

D


Gradient-Free Token Eviction for Edge-Deployable Video LLMs in Ecological Monitoring

Motivation: the edge bottleneck

Key idea

Method

Results

19.1%

52%

4.5 GB

37

Takeaways

Limitations

E

Lalu: dari paper ke kelas

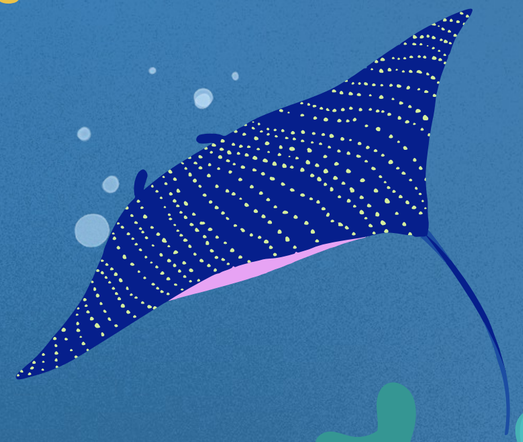
Istirahat

5 menit



ML dan Optimasi





Fungsi tujuan

Apa yang ingin diperkecil atau diperbesar.

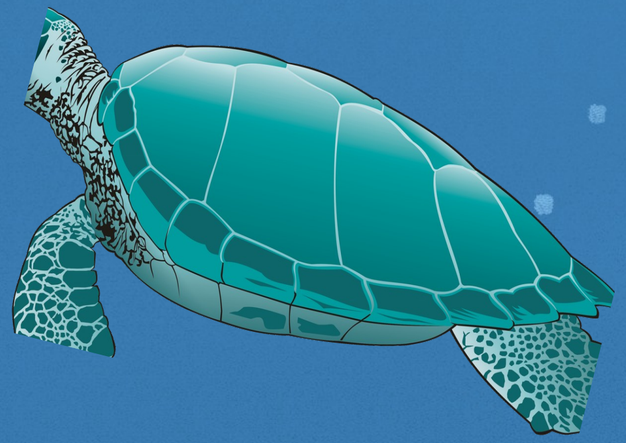
Ruang pencarian

Pilihan apa saja yang boleh kita coba.

Anggaran

Berapa kali kita boleh mencoba.

Evolusi Gambar



Satu aturan

Ubah sedikit. Simpan jika lebih mirip target, buang jika tidak.



Perhatikan

Kapan kurva turun cepat? Kapan melambat?

Ritme Mingguan

Sekitar 5 sampai 6 jam per pekan

1. Baca

Bab buku dan satu paper, dengan panduan baca.

2. Kelas

2 jam tatap muka: kuis tanpa perangkat, contoh, latihan berpasangan.

3. Latihan

Notebook Colab dan satu pertanyaan pemahaman.



Buku untuk Dasar, Paper untuk Garis Depan

Dua buku, gratis dan sah

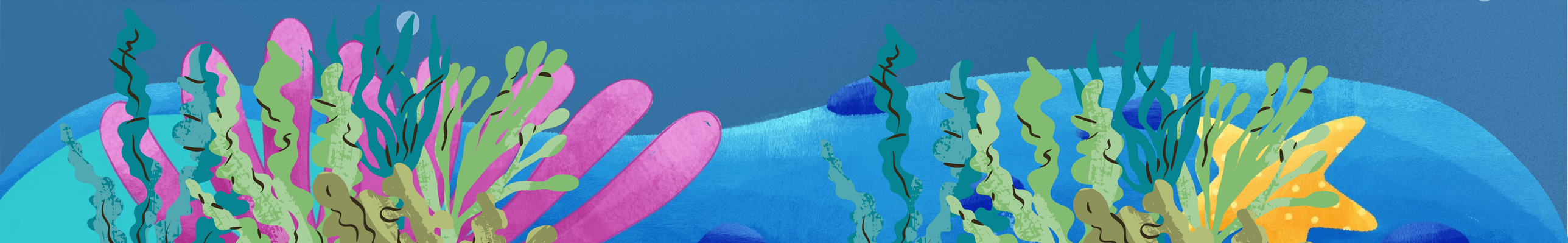
Understanding Deep Learning
Simon J. D. Prince, MIT Press 2023

Essentials of Metaheuristics
Sean Luke, edisi kedua, versi 2.4

Unduh dari situs penulis (Panduan Peserta)

Paper yang kita baca bersama

- Pekan 7: audit bias warna kulit
- Pekan 9 sampai 11: KMA dan penerusnya
- Pekan 12: token eviction dengan BPSO
- Pekan 14: CoT dan PoT pada model kecil



Penilaian

40%

Latihan

mingguan: kelengkapan dan satu pertanyaan pemahaman

10%

Cek lisan

Pekan 8 individu, Pekan 12 tim

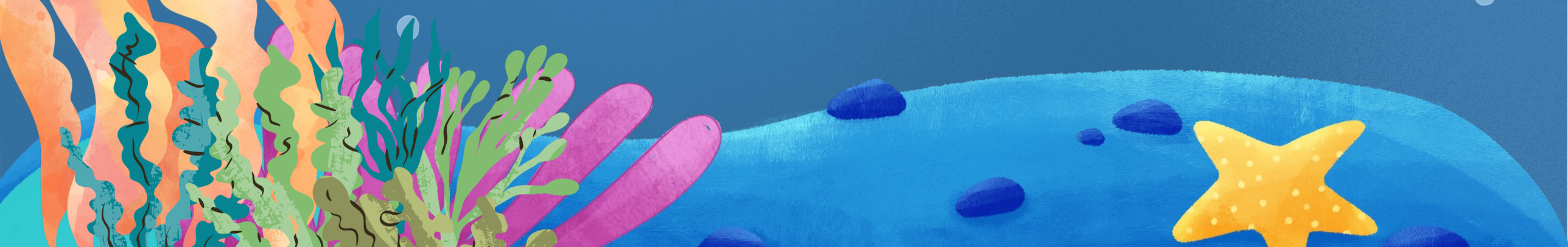
50%

Proyek akhir

tim 30%, cek lisan individu 20%

Kehadiran menjadi syarat: minimal 13 dari 16 pertemuan.





Proyek Akhir

- Tim 3 orang: model ML dengan lapisan optimasi yang menentukan hasil
- Dibandingkan dengan pembanding pada anggaran sama, minimal 10 percobaan
- **Catatan teknis 2 sampai 3 halaman: paper mini dengan satu kasus gagal**
- **Hari demo: sesi poster seperti konferensi**

Proposal Pekan 9 · Baseline Pekan 11 · Demo Pekan 16

Alat AI

Pakai AI apa pun, kapan pun. Tidak ada yang mencatatnya.

Tiga hal tetap berlaku:

1. Kuis pembuka dan cek lisan: tanpa perangkat
2. Angka yang kamu kutip dari paper dicek ke paper aslinya
3. Kamu yang bertanggung jawab dan harus bisa menjelaskannya

Stasiun Persiapan

A · Google dan Colab

Akun Google masuk. Colab terbuka dan print("halo") berjalan.

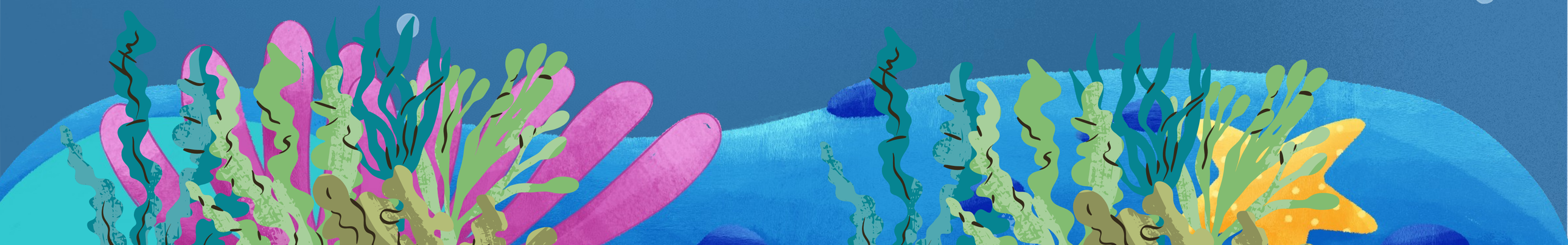
B · GitHub

Akun GitHub dibuat. Username diberikan ke mentor.

C · Buku

Kedua buku terunduh dan terbuka di perangkatmu.

10 menit. Yang belum selesai dikerjakan di rumah.



Untuk Pertemuan 1

- Baca UDL bagian 1.1 (halaman 1 sampai 6) dan bab 2 (halaman 17 sampai 22)
- Kerjakan lembar kerja Python untuk programmer Go (latihan 1 sampai 10)
- Bawa laptop yang sudah bisa membuka Colab

Tiket keluar (2 menit)

Tulis satu pertanyaan dan satu hal yang kamu khawatirkan.

Advance Study Group

Terima Kasih

Sampai jumpa

